

Hospital Risk Intelligence API - Deployment Guide

Prerequisites

- Python 3.10+
 - Docker Desktop
 - Windows/Linux/macOS
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Local Deployment

Step 1: Clone Project

```
git clone <repository_url>  
cd Capstone_Project
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Run API

```
uvicorn Phase5_APIIntegration.main:app --reload
```

Step 4: Access API

Home Page:

<http://localhost:8000>

Swagger Documentation:

<http://localhost:8000/docs>

Docker Deployment

Build Docker Image

```
docker build -t hospital-risk-api:1.0.0 .
```

Run Docker Container

```
docker run -d --name hospital-api -p 8000:8000 hospital-risk-api:1.0.0
```

Verify Running Container

```
docker ps
```

Expected:

Container status should be:

Up (healthy)

Health Check

```
GET /health
```

Expected Response:

```
{
  "status": "healthy",
  "model_version": "1.0.0"
}
```